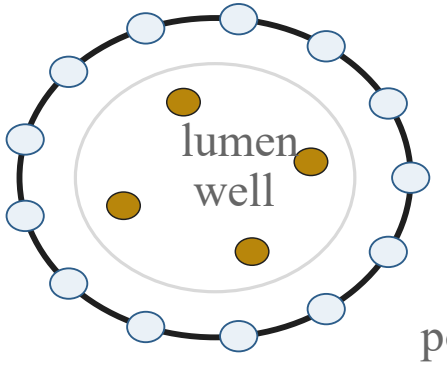
Candidate Microtubule Quantum-Well Scenario

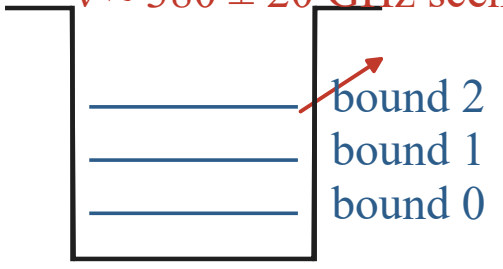
Microtubule Cross-Section Quantum Well



Trp sites in lumen wall

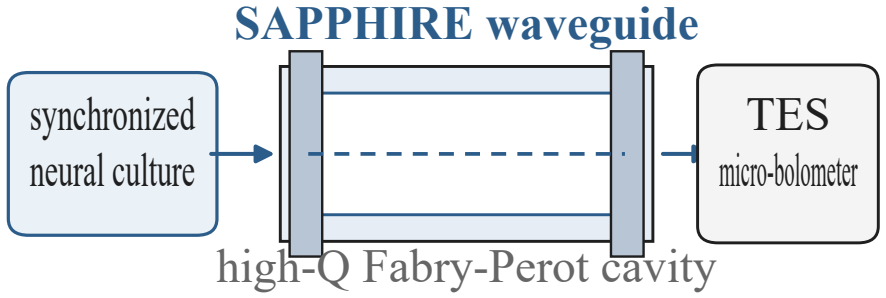
hypothetical leakage

$\nu \approx 380 \pm 20$ GHz scenario [Tier 3]



potential well diagram

Detection Apparatus



high-Q Fabry-Perot cavity

time-binned against MEA/gamma-burst window

Material-proxy test (O.10)

anaesthetic-modulation differential; no decisive gate
historical $N \gtrsim 10^5$ design estimate; detection conditional on
cryogenic TES + high-Q upgrade

after Dicke integration: ~500× below the cavity thermal floor
(multimode) to SNR ~6 (mode-matched); bracket-dependent, O.10